

Angular resolution enhancement technique for diffusion-weighted imaging (DWI) using predicted diffusion gradient directions

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ABSTRACT

Anisotropic diffusion MRI techniques using single-shell or multi-shell acquisitions have been proposed as a means to overcome some limitations imposed by diffusion tensor imaging (DTI), especially in complex models of fibre orientation distribution in voxels. A long acquisition time for the angular resolution of diffusion MRI is a major obstacle to practical clinical implementations. In this paper, we propose a novel method to improve angular resolution of diffusion MRI acquisition using given diffusion gradient (DG) directions. First, we define a local diffusion pattern map of diffusion MR signals on a single shell in given DG directions. Using the local diffusion pattern map, we design a prediction scheme to determine the best DG direction to be synthesized within a nearest neighborhood DG directions group. Second, the local diffusion pattern map and the spherical distance on the shell are combined to determine a synthesized diffusion signal in the new DG direction. Using the synthesized and measured diffusion signals on a single sphere, we estimate a spin orientation distribution function (SDF) with human brain data. Although the proposed method is applied to SDF, a basic idea is to increase the angular resolution using the measured diffusion signals in various DG directions. The method can be applicable to different acquired multi-shell data or diffusion spectroscopic imaging (DSI) data. We validate the proposed method by comparing the recovered SDFs using the angular resolution enhanced diffusion signals with the recovered SDF using the measured diffusion data. The developed method provides an enhanced SDF resolution and improved multiple fiber structure by incorporating synthesized signals. The proposed method was also applied neurite orientation dispersion and density imaging (NODDI) using multi-shell acquisitions.

1. Introduction

Diffusion tensor imaging (DTI) is a promising non-invasive magnetic resonance imaging (MRI) technique to visualize and characterize the diffusion properties of water molecules in the brain ((LeBihan et al., 1986; Merboldt et al., 1985; Taylor and Bushell, 1985; Zhang et al., 2011; Neil et al., 1998)). Combining at least 6 independent diffusion gradient (DG) directions, the measurement of the water diffusion tensor within each voxel of MRI provides insights into the microstructure of the brain including the orientational architecture of neural tissue and brain network connectivity ((Basser et al., 1994)). The diffusion tensor characterizes the path of white matter fiber bundles in the brain ((Hagmann et al., 2006)). DTI has been used to detect structural differences and to provide an indication of disease status in a range of neuropsychiatric and

neurological disorders; multiple sclerosis ((Sigal et al., 2012)), Huntington's disease ((Rosas et al., 2010)), schizophrenia ((Manoach et al., 2007)), obsessive-compulsive disorder ((Cannistraro et al., 2007)), bipolar disorder ((Beyer et al., 2005)), Alzheimers disease ((Gold et al., 2010; Pievani et al., 2010)), and amnesic mild cognitive impairment ((Agosta et al., 2011)).

Although DTI is a currently popular technique to visualize major fiber orientation in the brain, in fitting a Gaussian distribution of diffusion, DTI has a limitation in characterizing the diffusion process in areas of low anisotropy and complex fiber structure in a voxel. The high angular resolution diffusion imaging (HARDI) technique has been introduced to overcome the limitation of DTI ((Tuch et al., 2002)). To provide a better three dimensional architecture of neural tissue, HARDI uses more gradient encoding directions and high gradient strength (*b*-value) than

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DTI which uses fewer gradient directions at relatively lower b -values. The q -space diffusion imaging provides the empirical distribution of diffusion and estimates a three-dimensional (3D) probability density function of diffusion displacement. Several q -space reconstruction methods have been proposed to reconstruct the diffusion orientation distribution function (ODF) from diffusion MR signals. Q-ball imaging (QBI), one of the HARDI methods, directly derives ODF of water molecules and this ODF can be estimated directly from the raw HARDI signal S on a single sphere of q -space ((Tuch, 2004)). Based on the Funk-Radon transform (FRT), the FRT value at a given spherical point is a great circle integral of the signal on the sphere defined by the plane through the origin perpendicular to the point of evaluation. By investigating the Fourier transform relation between the diffusion MR signals and the diffusion displacement of the spins, a generalized q -sampling imaging method (GQI) has been proposed ((Yeh et al., 2010)). GQI is a model-free method that calculates the spin distribution function (SDF) directly from q -space which is the density of diffusing proton spins. If we normalize the SDF, it turns into the diffusion ODF. GQI has been shown to improve the quality of the crossing fibers if there is enough diffusion sampling and a recent comparison study has shown that GQI method offers high validation connections ((Maier-Hein et al., 2017)). A wide number of HARDI approaches have been proposed to estimate the underlying fiber orientation distributions including diffusion spectrum imaging (DSI) ((Wedeen et al., 2008)) and constrained spherical deconvolution (CSD) ((Mormina et al., 2015)).

In practical terms, to estimate reliable noninvasive brain diffusivity properties capable of revealing such white matter alteration patterns, a serious problem with these methods is a long data acquisition time due to many angular directions, which makes them difficult to routinely use for practical clinical applications ((Zhan et al., 2008)). Acquiring images with higher resolution increase the precision of anisotropy, but there is a trade-off between angular resolution and MR scan time.

To overcome the difficulties of long acquisition time, there have been several works by spatial interpolation methods based on neighbor linear interpolations ((Cho et al., 2009; Alaya et al., 2017; Bates et al., 2016)). Compressive sensing (CS) and sparse regularization techniques were proposed to reduce the MR data acquisition time. The CS technique in HARDI was used to construct highly sparse representation using the basis of spherical ridgelets ((Michailovich and Rath, 2010; Rath et al., 2014)). Scattered data interpolation methods, which have been an active area of research in statistics and many fields of engineering during the past half of a century, have been applied to MRI to increase the spatial resolution ((Peled and Yeshurun, 2002; Van Reeth et al., 2012)). To improve the coherence of q -ball ODFs, the non-local means filter was developed and produced more accurate ODF fields ((Descoteaux et al., 2008)). Recently, under the assumption that MR images are collected using a common imaging protocol, the angular resolution enhancement was developed to improve ODF estimation ((Chen et al., 2016)). However, it is still difficult to improve the resolution of diffusion characteristics from HARDI signals using the linear interpolation methods without additional information.

In this paper, we develop a new method to extract directional and quantitative information about the crossing fibers by enhancing the angular resolution of diffusion signals from a relatively small number of DG directions comparing to the conventional diffusion MRI acquisition. A key motivation is that a local diffusion pattern map of raw HARDI signals in given gradient directions can predict an appropriate direction and synthesize HARDI signal at the predicted directions. Since the anisotropic water diffusions in the nervous system are related to white-matter connections correlated with specific cognitive functions, the diffusion patterns in a set of adjacent GD directions are related to a spatial intermediate diffusivity value. For similar diffusion patterns in the adjacent GD directions, we can predict a proper DG direction as an appropriate combination of adjacent DG directions. The local diffusion pattern maps as a characteristic map of raw HARDI signals and the spherical distances are combined to synthesize a HARDI signal at a new DG direction. We predict a new DG direction by the weighted

combination of measured positions in the q -space on a single sphere. We use a correlation coefficient between the behavior of diffusion signals in a local window as the distance of local diffusion patterns. In the new DG direction, the difference between the synthesized diffusion signal and the actual unmeasured diffusion signal depends on the local diffusion pattern and measured noise artifact. The noise variance of the synthesized diffusion signal in the new direction is less than the noise level of diffusion signals at each measured position on the sphere. The synthesized diffusion signal at the predicted position on the sphere can approximately recover the actual non-measured diffusion signal by analyzing the local diffusion pattern map.

Using the measured diffusion signals on a spherical shell in q -space, we estimate the SDFs using the GQI technique to resolve intravoxel white matter fiber heterogeneity. We acquired human brain data using 64 DG directions with a b -value of 2000 s/mm². To validate the proposed method, we compared the recovered SDFs using the synthesized and measured diffusion signals in 32 DG directions with the recovered SDF using the measured data in 64 DG directions. We compared the relative errors of SDFs between 64 DG directions and the predicted DG directions in the brain region. The proposed enhanced angular resolution of diffusion MR acquisitions implies that any previously scanned HARDI data set can be reused to improve SDF quality for intravoxel white matter fiber heterogeneity without additional experiments. For another experiment, using publicly available human connectome project (HCP) data, we also estimated and compared the qualities of SDFs.

Although we use q -ball imaging and HARDI to explain how the proposed method increases the angular resolution, the process for predicting and synthesizing diffusion signals in a new DG direction can be applied to different acquisition methods including multi-shell diffusion imaging and diffusion spectrum magnetic resonance imaging (DSI) as well as different reconstruction methods (constrained spherical harmonics (CSD), ball-and-sticks, neurite orientation dispersion and density imaging (NODDI), etc). We applied the proposed method to multi-shell data and reconstructed the neurite density and orientation dispersion using NODDI. In [Supplementary material](#), we compared the results.

2. Theory

2.1. q -space and ensemble average propagator (EAP)

The pulsed gradient spin echo (PGSE) excites the spin system by applying time-constant magnetic field gradients before and after the π -pulse and acquires the k -space data. Due to the dephasing of magnetization because of diffusion in the period Δ , the acquired signal intensity reflects the amount of diffusion occurred during the diffusion gradient pulses. The diffusion signal can be represented as

$$S(\mathbf{g}, \Delta, \delta) = \int_{\mathbb{R}^3} \rho(\mathbf{R}_0) \int_{\mathbb{R}^3} P(\mathbf{R}_\Delta | \mathbf{R}_0) \exp(i\gamma\delta(\mathbf{R}_\Delta - \mathbf{R}_0) \cdot \mathbf{g}) d\mathbf{R}_\Delta d\mathbf{R}_0,$$

where $\mathbf{g}$ is the diffusion-sensitizing gradient vector, δ is the diffusion gradient duration, $\gamma = 26.75 \times 10^7$ rad/Ts is gyromagnetic ratio, $\rho(\mathbf{R}_0)$ is the spin density at the initial time $t = 0$ and $P(\mathbf{R}_\Delta | \mathbf{R}_0)$ is the probability that the spin moves from $\mathbf{R}_0$ at $t = 0$ to $\mathbf{R}_\Delta$ at $t = \Delta$. Using the ensemble average propagator (EAP) as

$$P(\mathbf{R}) = \int_{\mathbb{R}^3} \rho(\mathbf{R}_0) P(\mathbf{R}_0 + \mathbf{R} | \mathbf{R}_0) d\mathbf{R}_0, \quad \mathbf{R} = \mathbf{R}_\Delta - \mathbf{R}_0,$$

The signal attenuation is represented by the inverse Fourier transform of EAP as

$$S(\mathbf{g}) = \int_{\mathbb{R}^3} P(\mathbf{R}) \exp(2\pi i \mathbf{q} \cdot \mathbf{R}) d\mathbf{R},$$

where $\mathbf{q} = \frac{\gamma\delta\mathbf{g}}{2\pi}$ is a vector in q -space ((Stejskal and Tanner, 1965)). The diffusion weighting factor, b -value, is represented by

$$b = \gamma^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right) \|\mathbf{g}\|^2.$$

The EAP is assumed to be radially symmetric, i.e. $P(\mathbf{R}) = P(-\mathbf{R})$. Diffusion in the x -, y -, and z -directions can be measured by applying the gradient in the x -, y -, and z -directions, respectively.

A typical HARDI signal is obtained from spherical sampling, and so a spherical basis is an appropriate representation for a raw HARDI signal. The HARDI requires a finer angular resolution sampling comparing to the conventional DTI sampling scheme. The HARDI methods estimate ODF in multiple shell or single shell sampling data. In spite of principal advantage of q -space approaches, the main disadvantage is that the HARDI technique requires a large number of measurements to encode q -space and thus it takes a long acquisition time, restricting its use in the clinical implementation.

2.2. Prediction of diffusion gradient direction using a local diffusion pattern map

For a given voxel coordinate $\mathbf{r}$ in an imaging region and a set $\mathcal{S}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_N\}$ of DG directions, let $S(\mathbf{g}_i) = S(\mathbf{g}_i, \mathbf{r})$, $i = 1, \dots, N$, be the measured diffusion MR signals for the DG directions $\mathcal{S}_N$ and a fixed b -value.

In this paper, we aim to generate appropriate diffusion MR signals on a finer set $\mathcal{S}_M$ of DG directions including the DG directions $\mathcal{S}_N$. We denote the elements of $\mathcal{S}_M$ as $\mathcal{S}_M = \{\mathbf{g}_1, \dots, \mathbf{g}_N, \mathbf{v}_1, \dots, \mathbf{v}_{M-N}\}$. Instead of a simple approximation method, we shall generate diffusion MR signals, $S(\mathbf{v}_j)$, $j = 1, \dots, M - N$, by taking into account the similarity of the measured MR signals $S(\mathbf{g}_i)$ in neighborhood DG directions. For a direction $\mathbf{v}_j \in \mathcal{S}_M \setminus \mathcal{S}_N$, let $\mathcal{N}_{\mathbf{v}_j}^K = \{\mathbf{g}_{j_k} \in \mathcal{S}_N : k = 1, \dots, K\}$ be the set of K directions $\mathbf{g}_{j_k}$ close to $\mathbf{v}_j$.

A basic motivation is that local diffusion pattern similarities in the neighborhood $\mathcal{N}_{\mathbf{v}_j}^K$ are related to the diffusion signals in the convex hull, $\text{Conv}(\{\mathbf{g}_{j_1}, \dots, \mathbf{g}_{j_K}\}) = \{\sum_{k=1}^K w_k \mathbf{g}_{j_k} : \sum_{k=1}^K w_k = 1, w_k \geq 0\}$:

$$p_n(S(\mathbf{g}_{j_m})) \approx p_n(S(\mathbf{g}_{j_n})) \text{ for } m, n = 1, \dots, K \text{ implies} \quad (1)$$

$$S(\mathbf{u}_c) := S\left(\sum_{k=1}^K w_k \mathbf{g}_{j_k}\right) \approx \sum_{k=1}^K w_k S(\mathbf{g}_{j_k}),$$

where $p_n(S(\mathbf{g}_{j_k}))$ denotes a local diffusion pattern map of the diffusion signal $S(\mathbf{g}_{j_k})$ and $\mathbf{u}_c \in \text{Conv}(\{\mathbf{g}_{j_1}, \dots, \mathbf{g}_{j_K}\})$. Depending on the similarity between the local diffusion patterns in $\mathcal{N}_{\mathbf{v}_j}^K$, we determine a new appropriate direction

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{j_k} \quad (2)$$

in $\text{Conv}(\{\mathbf{g}_{j_1}, \dots, \mathbf{g}_{j_K}\})$, which is synthesized by the measured directions in the neighborhood $\mathcal{N}_{\mathbf{v}_j}^K \subset \mathcal{S}_N$. In this paper, we take a local diffusion pattern map, p_n in (1), at a spatial voxel $\mathbf{r} = (x, y, z)$ between two diffusion MR signals $S(\mathbf{g}_{j_m})$ and $S(\mathbf{g}_{j_n})$ as a correlation coefficient map using a $3 \times 3 \times 3$ local window $\omega_r = \{(x + l_x, y + l_y, z + l_z) : l_x, l_y, l_z = -1, 0, 1\}$ centered at $\mathbf{r}$. Let $S_{j_m} = \{S(\mathbf{g}_{j_m}, (x + l_x, y + l_y, z + l_z)) : l_x, l_y, l_z = -1, 0, 1\}$ be the set of diffusion MR signals around $S(\mathbf{g}_{j_m})$ in the local window ω_r . The correlation coefficient $CC(S_{j_m}, S_{j_n})$ between two set of neighboring diffusion MR signals S_{j_m} and S_{j_n} is defined as

$$CC(S_{j_m}, S_{j_n}) = \frac{\text{Cov}(S_{j_m}, S_{j_n})}{\sqrt{\text{Cov}(S_{j_m}, S_{j_m}) \text{Cov}(S_{j_n}, S_{j_n})}} \quad (3)$$

where

$$\text{Cov}(S_{j_m}, S_{j_n}) = \frac{1}{27} \sum_{l_x, l_y, l_z = -1}^1 \left(S(\mathbf{g}_{j_m}, (x + l_x, y + l_y, z + l_z)) - \bar{S}_{j_m} \right) \times \left(S(\mathbf{g}_{j_n}, (x + l_x, y + l_y, z + l_z)) - \bar{S}_{j_n} \right)$$

Here, $\bar{S}_{j_m}$ is the average of MR signals in S_{j_m} .

To determine the weighting factors w_k , $k = 1, \dots, K$ in (2), we use the correlation coefficient $CC(S_{j_m}, S_{j_n})$ as a local diffusion pattern map for $S(\mathbf{g}_{j_m})$ and $S(\mathbf{g}_{j_n})$. Define

$$\mathbf{A} = \begin{bmatrix} a_{11} & \dots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{KK} \end{bmatrix}$$

where

$$a_{mn} = \begin{cases} CC(S_{j_m}, S_{j_n}), & m \neq n \\ 0, & m = n. \end{cases}$$

For the measured DG direction $\mathbf{g}_{j_k}$, $j \in \{1, \dots, K\}$, we set a similarity weight as

$$w_k = \sum_{n=1}^K a_{kn} / \|\mathbf{A}\|$$

where $\|\mathbf{A}\| = \sum_{m=1}^K \sum_{n=1}^K a_{mn}$. Using these weights, we determine a new predicted DG direction

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{j_k} = \sum_{k=1}^K \sum_{n=1}^K a_{kn} / \|\mathbf{A}\| \mathbf{g}_{j_k} \in \text{Conv}(\mathcal{N}_{\mathbf{v}_j}^K). \quad (4)$$

This direction $\mathbf{u}_c$ satisfies a balancing condition

$$\frac{1}{\|\mathbf{A}\|} \sum_{m=1}^K \sum_{n=1}^K a_{mn} (\mathbf{g}_{j_k} - \mathbf{u}_c) = 0.$$

The proposed method to predict a DG direction uses the diffusion similarity and determines the DG direction using a linear combination of the surrounding DG directions.

To specify the proposed method and to select more predicted directions, we first select the four nearest directions $\{\mathbf{g}_{j_k} \in \mathcal{S}_N : k = 1, \dots, 4\}$ around $\mathbf{v}_j \in \mathcal{S}_M \setminus \mathcal{S}_N$ and then select the four triangles as shown in Fig. 1(a). One of the selected triangles is denoted by $\mathcal{N}_{\mathbf{v}_j}^{3,1}$.

The correlation coefficients in the local neighborhood $\mathcal{N}_{\mathbf{v}_j}^{3,1}$ determine a location as an intermediate DG direction. For the local neighborhood $\mathcal{N}_{\mathbf{v}_j}^{3,1} = \{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}\}$, we predict an appropriate DG direction, $\mathbf{u}_{c_1}$, only when the correlation coefficient provides a positive relation; $CC(S_{j_1}, S_{j_2}) > 0$, $CC(S_{j_2}, S_{j_3}) > 0$, and $CC(S_{j_3}, S_{j_1}) > 0$. In that case, a new predicted DG direction $\mathbf{u}_{c_1}$ is determined as follows:

$$\mathbf{u}_{c_1} = \sum_{k=1}^3 w_k \mathbf{g}_{j_k}$$

where

$$w_1 = \frac{CC(S_{j_1}, S_{j_2}) + CC(S_{j_3}, S_{j_1})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))},$$

$$w_2 = \frac{CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))},$$

$$\text{and } w_3 = \frac{CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1})}{2(CC(S_{j_1}, S_{j_2}) + CC(S_{j_2}, S_{j_3}) + CC(S_{j_3}, S_{j_1}))}.$$

Repeating this procedure for all triangles $\mathcal{N}_{\mathbf{v}_j}^3$ in $\{\mathbf{g}_{j_k} \in \mathcal{S}_N : k = 1, \dots, 4\}$, we can include up to four predicted positions. As an example, Fig. 1(a) represents the selected DG directions $\{\mathbf{g}_{j_k} \in \mathcal{S}_N : k = 1, \dots, 4\}$

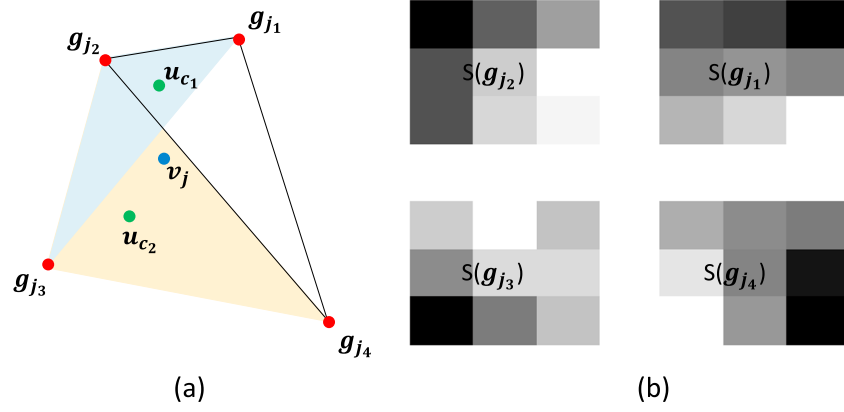


Fig. 1. Diagram for predicting directions in $\text{Conv}(\{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}, \mathbf{g}_{j_4}\})$. (a) Selected DG directions $\{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}, \mathbf{g}_{j_4}\}$ and determined predicted directions $\mathbf{u}_{c_1}$ and $\mathbf{u}_{c_2}$ within triangles $\mathcal{N}_{v_j}^{3,1} = \{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}\}$ and $\mathcal{N}_{v_j}^{3,2} = \{\mathbf{g}_{j_2}, \mathbf{g}_{j_3}, \mathbf{g}_{j_4}\}$, respectively. (b) Intensity of $S(\mathbf{g}_{j_k}, (\mathbf{x} + l_x, \mathbf{y} + l_y, \mathbf{z} + l_z))$, $l_x, l_y = -1, 0, 1$, $l_z = 0$, $k = 1, \dots, 4$, in a $3 \times 3 \times 3$ local window.

and determined predicted positions $\mathbf{u}_{c_1}$ and $\mathbf{u}_{c_2}$ within triangles $\mathcal{N}_{v_j}^{3,1} = \{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}\}$ and $\mathcal{N}_{v_j}^{3,2} = \{\mathbf{g}_{j_2}, \mathbf{g}_{j_3}, \mathbf{g}_{j_4}\}$, respectively. Fig. 1(b) shows a portion of MR signals in a $3 \times 3 \times 3$ local window in given DG directions. The diffusion patterns are used to calculate the weighting factors w_k within the triangle neighborhoods $\mathcal{N}_{v_j}^3$.

2.3. Synthesized MR diffusion signal at a predicted gradient direction

The weighted combination of the correlation coefficients $CC(S_{j_m}, S_{j_n})$ between two set of neighboring diffusion MR signals S_{j_m} and S_{j_n} determines the location $\mathbf{u}_c$ which is a convex combination of DG directions in $\mathcal{N}_{v_j}^K$. Since the weighting factors w_k for the predicted DG direction, $\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{j_k}$ in (4), is determined with respect to the diffusion similarity, using the same weighting factors, we generate a synthesized MR diffusion signal at $\mathbf{u}_c$ as

$$S(\mathbf{u}_c) = S\left(\sum_{k=1}^K w_k \mathbf{g}_{j_k}\right) = \sum_{k=1}^K w_k S(\mathbf{g}_{j_k}) \quad (5)$$

For the unmeasured true MR diffusion signal, $S^r(\mathbf{u}_c)$, at $\mathbf{u}_c \in \text{Conv}(\mathcal{N}_{v_j}^K)$ and the estimated $S(\mathbf{u}_c)$, the error between $S^r(\mathbf{u}_c)$ and $S(\mathbf{u}_c)$ can be decomposed into the following form:

$$\begin{aligned} |S^r(\mathbf{u}_c) - S(\mathbf{u}_c)| &= \left| S^r(\mathbf{u}_c) - \sum_{k=1}^K w_k S^r(\mathbf{g}_{j_k}) + \sum_{k=1}^K w_k (S^r(\mathbf{g}_{j_k}) - S(\mathbf{g}_{j_k})) \right| \\ &\leq \left| S^r(\mathbf{u}_c) - \sum_{k=1}^K w_k S^r(\mathbf{g}_{j_k}) \right| + \left| \sum_{k=1}^K w_k (S^r(\mathbf{g}_{j_k}) - S(\mathbf{g}_{j_k})) \right| \end{aligned}$$

The first term on the right side of the last expression shows the reliability error from the prediction model. We can express the reliability error in the following form:

$$\left| S^r(\mathbf{u}_c) - \sum_{k=1}^K w_k S^r(\mathbf{g}_{j_k}) \right| = \left| S^r\left(\sum_{k=1}^K w_k \mathbf{g}_{j_k}\right) - \sum_{k=1}^K w_k S^r(\mathbf{g}_{j_k}) \right| \quad (6)$$

The relation (6) shows that the weighting factors w_k , $k = 1, \dots, K$, determined to predict the location $\mathbf{u}_c$, reflect the similarity of diffusion properties and can reduce the difference between $S^r(\mathbf{u}_c)$ and $\sum_{k=1}^K w_k S^r(\mathbf{g}_{j_k})$ depending on the local diffusion patterns similarity.

The second term, $\sum_{k=1}^K w_k (S^r(\mathbf{g}_{j_k}) - S(\mathbf{g}_{j_k}))$, represents the error term from the measured noises. For the second error term, since the noise variance of the synthesized diffusion MR signal at $\mathbf{u}_c$ is $\text{Var}(\sum_{k=1}^K w_k (S^r(\mathbf{g}_{j_k}) - S(\mathbf{g}_{j_k}))) = \sum_{k=1}^K w_k^2 \text{Var}(S^r(\mathbf{g}_{j_k}) - S(\mathbf{g}_{j_k}))$, the

noise level of the synthesized diffusion MR signal at the predicted location is less than the noise level of measured diffusion MR signals at each DG direction, assuming that the noise is independent and identically distributed.

2.4. High resolution MR diffusion signals

For a given voxel coordinate $\mathbf{r}$ and measured MR diffusion signals $S(\mathbf{g}_i)$, $i = 1, \dots, N$, we aim to generate appropriate diffusion MR signals on a finer set $\mathcal{S}_M = \{\mathbf{g}_1, \dots, \mathbf{g}_N, \mathbf{v}_1, \dots, \mathbf{v}_{M-N}\}$. Let $\mathcal{S}_C = \{\mathbf{g}_1, \dots, \mathbf{g}_N, \mathbf{u}_{c_1}, \dots, \mathbf{u}_{c_L}\}$ be the total set of measured diffusion gradient directions $\mathbf{g}_i$ and the synthesized directions $\mathbf{u}_{c_m}$ estimated by the diffusion pattern similarity in (4). We use the radial basis function (RBF) approximation to determine the MR signals $S(\mathbf{v}_j)$, $j = 1, \dots, M - N$. The MR diffusion signals $S(\mathbf{v}_j)$ are defined by

$$S(\mathbf{v}_j) = \sum_{\mathbf{g} \in \mathcal{S}_C} c_{\mathbf{g}} \phi(\mathbf{v}_j - \mathbf{g}), \quad j = 1, \dots, M - N, \quad (7)$$

where ϕ is a positive definite radial basis function and $c_{\mathbf{g}}$, $\mathbf{g} \in \mathcal{S}_C$, are the weight coefficients. In this paper, we use Gaussian function $\phi(\cdot) = e^{-a\|\cdot\|^2}$. To estimate the weight coefficients $c_{\mathbf{g}}$, we solve the following linear system,

$$(\mathbf{A} + \lambda \mathbf{I}) \mathbf{c} = \mathbf{b} \quad (8)$$

where

$$\mathbf{A} = \begin{bmatrix} e^{-a\|\mathbf{g}_1 - \mathbf{g}_1\|^2} & \dots & e^{-a\|\mathbf{g}_1 - \mathbf{u}_{c_L}\|^2} \\ \vdots & \ddots & \vdots \\ e^{-a\|\mathbf{u}_{c_L} - \mathbf{g}_1\|^2} & \dots & e^{-a\|\mathbf{u}_{c_L} - \mathbf{u}_{c_L}\|^2} \end{bmatrix},$$

$$\mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_{N+L} \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} S(\mathbf{g}_1) \\ \vdots \\ S(\mathbf{u}_{c_L}) \end{bmatrix}.$$

Here, λ is a regularization parameter and $\mathbf{I}$ denotes the identity matrix. The resulting weight coefficients vector $\mathbf{c}$ are the solution of the following regularized least squares approximation problem (Fasshauer, 2007):

$$\min_{\mathbf{c} \in \mathbb{R}^{N+L}} \left\{ (\mathbf{A}\mathbf{c} - \mathbf{b})^T (\mathbf{A}\mathbf{c} - \mathbf{b}) + \frac{\lambda}{2} \mathbf{c}^T \mathbf{A} \mathbf{c} \right\} \quad (9)$$

The regularization parameter λ controls the balance between the smoothness of interpolation function and the data fitting. The implementation procedure of the proposed algorithm is summarized as follows:

1. Determine the predicted diffusion gradient directions, $\mathbf{u}_{cm} \in \mathcal{D}_C$, depending on the diffusion similarity, $CC(S_{j_m}, S_{j_n})$, as in (3).
2. Generate a synthesized diffusion MR signal $S(\mathbf{u}_{cm})$, $\mathbf{u}_{cm} \in \mathcal{D}_C$, by combining measured signal $S(\mathbf{g}_j)$, $\mathbf{g}_j \in \mathcal{D}_N$, using the weighting factors which are determined by the relationship between $S(\mathbf{g}_j)$, within a local window.
3. Increase the diffusion gradient directions from the combined DG $\mathcal{D}_C$ to the fine DG set $\mathcal{D}_M$ by determining the weighting factor vector $\mathbf{c}$ by solving the matrix system (8).

3. Methods

3.1. Generalized q-sampling imaging

In this paper, we use the generalized q-sampling imaging (GQI) method to verify the proposed method. Generalized q-sampling imaging ((Yeh et al., 2010)), one of the q -space imaging methods, is based on an analytical relation between diffusion signals and the spin distribution function (SDF) Ψ , which is the density of diffusing water at different direction:

$$\Psi(\mathbf{u}) = z_0 \sum_{\mathbf{g}} S(\mathbf{g}) \text{sinc}(\sigma \sqrt{6Db} \mathbf{g} \cdot \mathbf{u}), \quad (10)$$

where z_0 is a constant that unifies spin density with respect to the amount of water, σ is a sampling length ratio, D is the self-diffusion coefficient of water at the body temperature. GQI overcomes the difficulties of DTI, which has the problem of resolving crossing fibers, and is advantageous over DTI for the tractography of the fiber bundle. The SDF is a non-weighted version of the diffusion oriented distribution function demonstrated by q -ball imaging (QBI) or diffusion spectrum imaging (DSI). If we normalize the SDF then it turns into the diffusion ODF.

The GQI method can be applied to the shell sampling scheme and can provide directional and quantitative information about the crossing fibers. The GQI is good for tractography if there is enough diffusion sampling and a recent comparison study has shown that GQI method offers high validation connections ((Maier-Hein et al., 2017)).

In this paper, SDFs were calculated using DSI Studio software (<http://dsi-studio.labsolver.org>), an open-source diffusion MRI analysis toolbox. DSI studio is an open popular diffusion MRI analysis tool that maps brain connections and correlates findings with neuropsychological disorders ((Yeh et al., 2017; Kearney-Ramos et al., 2018; Celtikci et al., 2018)).

3.2. Experimental setup

Two healthy volunteers without documented history of any disease were recruited. The participants were located inside the bore of a 3 T MRI scanner with an 32 channel RF head coil (Ingenia, Philips Medical Systems, the Netherlands). All experimental protocols were approved by the institutional review board of Kyung Hee University (KHSIRB-16-033). All methods were carried out in accordance with the relevant guidelines and regulations.

Case 1: Diffusion weighted imaging data sets were obtained using the single-shot spin-echo echo-planar-imaging pulse sequence. We applied the diffusion weighting gradients with b value of 2000 s/mm² and 64 DG directions. Imaging parameters were as follows: repetition time $T_R = 12000$ ms, echo time $T_E = 70$ ms, slice thickness 2 mm, field of view 224×224 , flip angle 90° , number of excitations 1, number of slices 55, and acquisition matrix 112×112 . The matrix size of 112×112 was reconstructed to 224×224 for subsequent data processing steps. The total scan time was 804 s.

Case 2: Diffusion weighted imaging data ($b = 2000$ s/mm² with 32 DG directions) were acquired using the single-shot spin-echo echo-planar-imaging pulse sequence with $T_R = 15000$ ms, $T_E = 80$ ms, slice thickness 2 mm, field of view 224×224 , flip angle 90° , number

of excitations 1, number of slices 54, and acquisition matrix 112×112 . The scan time was 525 s.

ExploreDTI tool was used to correct DWI images to reduce motion, eddy-current-induced geometric distortion artifacts (<http://www.exploredti.com>).

Moreover, we validated the proposed method using a single subject (100307) of the Human Connectome Project (HCP) diffusion weighted imaging data ((Van Essen et al., 2013)). The HCP data consisted of 270 diffusion weighted images distributed equally over 3 shells defined with b values of 1000 s/mm², 2000 s/mm², and 3000 s/mm². In each volume, acquisition matrix size was 145×174 , number of slices 145, and slice thickness 1.25 mm. We applied only shell with b value of 3000 s/mm² to the proposed method and the results are available in [Supplementary material](#).

3.3. Experimental validation

For experimentation procedure, the input diffusion MRI volume is of size $224 \times 224 \times 55$ and $\{\mathbf{g}_i\}_{i=1}^{64}$ are the 64 gradient directions. Fig. 2(a) shows the measured HARDI signals for 64 DG directions on the 30th imaging slice.

We estimated three kinds of SDFs. First, we found SDFs, denoted by f_{64} , using the full measured diffusion MR signals $\{S(\mathbf{g}_1), \dots, S(\mathbf{g}_{64})\}$ and compared the results, f_{64} , with the recovered SDF maps using a partial DG set to verify the proposed method. We used the upper 32 images in Fig. 2(a) as a partially measured HARDI signals with DG directions $\mathcal{D}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_{32}\}$ and corresponding SDFs are denoted by f_{32} .

The diffusion MR signals on DG set $\mathcal{D}_M \setminus \mathcal{D}_N = \{\mathbf{v}_1, \dots, \mathbf{v}_{32}\} = \{\mathbf{g}_{33}, \dots, \mathbf{g}_{64}\}$ are to be recovered using the proposed method. The DGs $\mathcal{D}_N$ are used to select a neighborhood $\mathcal{N}_{\mathbf{v}_j}^K$ for $\mathbf{v}_j \in \mathcal{D}_M \setminus \mathcal{D}_N$. Fig. 2(b) shows the DGs $\mathcal{D}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_{32}\}$ (red spots) and $\mathcal{D}_M \setminus \mathcal{D}_N = \{\mathbf{v}_1, \dots, \mathbf{v}_{32}\}$ (blue spots).

For each direction $\mathbf{v}_j \in \mathcal{D}_M \setminus \mathcal{D}_N$ at a voxel location $\mathbf{r}$, we selected four nearest DG directions $\{\mathbf{g}_k \in \mathcal{D}_N : k = 1, \dots, 4\}$. For a triangle neighborhood $\mathcal{N}_{\mathbf{v}_j}^3 = \{\mathbf{g}_{j_1}, \mathbf{g}_{j_2}, \mathbf{g}_{j_3}\}$, we determined a DG direction $\mathbf{u}_{c_1}$ only when $CC(S_{j_1}, S_{j_2}) > 0.01$, $CC(S_{j_2}, S_{j_3}) > 0.01$, and $CC(S_{j_3}, S_{j_1}) > 0.01$ as in Section 2.2. We continued in this fashion to predict new DG directions $\mathbf{u}_{cm}$ and synthesized corresponding diffusion MR signals $S(\mathbf{u}_{cm})$ at each voxel using (5). The predicted DG directions $\mathbf{u}_{cm}$ (green spots) at a voxel (92, 87, 30) were displayed in Fig. 2(b), where the number of predicted DG directions $\mathbf{u}_{cm}$ was 55. To move the DG directions $\mathcal{D}_C$ to $\mathcal{D}_M$, we solved the 87×87 size of matrix system $(\mathbf{A} + \lambda \mathbf{I})\mathbf{c} = \mathbf{b}$ in (8). To solve the weighted least square problem (9), we determined the regularization parameter λ using the generalized cross-validation (GCV) technique, which minimizes the predictive mean-square error without statistical information ((Hansen, 1992)).

With this optimal parameter λ , we used the singular value decomposition of $\mathbf{A}$ to find the solution of the regularized least squares problem (9). For the parameter value $\alpha = 1$, the determined regularization parameter λ was 0.0001.

Using the determined weighting vector $\mathbf{c}$, we estimated the MR diffusion signals $S(\mathbf{v}_j)$ using RBF approximation described in (7). We denoted by f_{cm} the reconstructed SDFs using $\{S(\mathbf{g}_1), \dots, S(\mathbf{g}_{32}), S(\mathbf{v}_1), \dots, S(\mathbf{v}_{32})\}$.

The sampling length ratio σ in (10) controls the displacement range of the diffusing spins sampled by the SDFs and serves as a regularization parameter to adjust the coarseness of the SDFs. The sampling length ratios σ were 0.8, 1 and 1 for f_{32} , f_{cm} and f_{64} , respectively.

The relative L^2 -error of a reconstructed normalized SDF $\tilde{f}_{32}$ and $\tilde{f}_{cm}$, comparing to $\tilde{f}_{64}$, is defined as:

$$RE_{32} := \frac{\|\tilde{f}_{64} - \tilde{f}_{32}\|}{\|\tilde{f}_{64}\|} \quad \text{and} \quad RE_{cm} := \frac{\|\tilde{f}_{64} - \tilde{f}_{cm}\|}{\|\tilde{f}_{64}\|}. \quad (11)$$

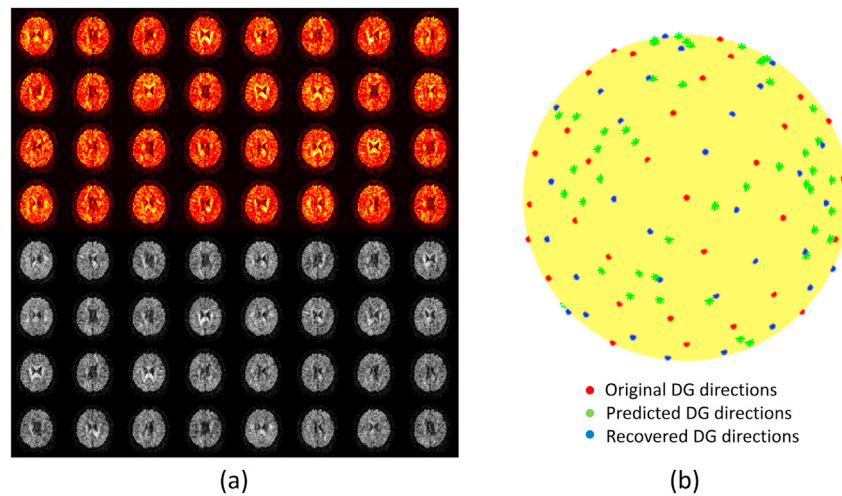


Fig. 2. (A) Measured HARDI signals for DG directions $\mathcal{S}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_{32}\}$ (upper 32 images). HARDI signals for $\{\mathbf{g}_1, \dots, \mathbf{g}_{64}\}$ are used to generate a target SDF distribution. (b) DG directions at a spatial position (92, 87, 30): $\mathcal{S}_N = \{\mathbf{g}_1, \dots, \mathbf{g}_{32}\}$ (red spots), $\mathcal{S}_M \setminus \mathcal{S}_N = \{\mathbf{v}_1, \dots, \mathbf{v}_{32}\}$ (blue spots), and predicted DG directions $\mathbf{u}_{cm}$ (green spots).

For the HCP data, we found SDFs, denoted by f_{90} , using the measured 90 diffusion MR signals. The first 30 signals were used as partially measured HARDI signals and the corresponding SDFs were denoted by f_{30} . We used the 30 diffusion MR signals to reconstruct new 60 diffusion MR signals and denoted the corresponding SDFs by f_{cm} . The sampling length ratios σ were 0.9, 0.7, and 0.85 for f_{90} , f_{30} , and f_{cm} , respectively.

4. Results

4.1. Experimental results for SDF

Fig. 3(a) shows the color-coded fractional anisotropy (FA) map which describes the degree of anisotropy by the diffusion tensor. For the diffusion tensor, we used the full measured diffusion MR signals $\{S(\mathbf{g}_1), \dots, S(\mathbf{g}_{64})\}$. The FA map only accounted for Gaussian diffusion processes, which was inadequate in accurately representing the intravoxel fiber crossing in the brain. In Fig. 3(b), three ROIs are marked by red, green, and blue boxes in the brain region. Fig. 3(c) shows the reconstructed SDFs using the full measured diffusion MR signals $\{S(\mathbf{g}_1), \dots, S(\mathbf{g}_{64})\}$, which provided a relatively extract directional and quantitative information about the crossing fibers. We compared the recovered SDFs, f_{64} , f_{32} , and f_{cm} , in the three ROIs (red, green and blue boxes), which enlarged along with the QBI glyphs in Fig. 3(d), (e), and (f), respectively. We observed that f_{cm} improved the shape of SDFs and that the volume size of SDFs was closer to f_{64} in most voxels (see also Supplementary Fig. 1).

The images of relative errors RE_{32} and RE_{cm} were shown in Fig. 4(b) and (c). We observed that RE_{cm} were much smaller than RE_{32} at most locations. Table 1 shows the reconstructed SDFs f_{64} , f_{32} , and f_{cm} , respectively, with resolved fiber orientations and two relative errors at different voxel locations. Fig. 4(a) shows the voxel locations in the brain region corresponding to the rows in Table 1. At the voxel locations (red spots) in Fig. 4(a), the SDF f_{cm} generated a multiple fiber structure more closely matched the SDF f_{64} , while the recovered SDF f_{32} missed the multiple fiber structure (see also Supplementary Fig. 2).

Fig. 5 shows the locations where the number of crossing-fibers for f_{64} , f_{32} , and f_{cm} , respectively. We observed that the crossing fibers were detected in 16%, 7%, and 17% of the brain area for f_{64} , f_{32} , and f_{cm} , respectively (see also Supplementary Fig. 3).

We evaluated the fiber structures by calculating the inner angles between the major fibers (F_{d1}) defined as the largest local maximum at SDFs f_{64} and f_{32} (or f_{cm}). The second fiber direction (F_{d2}) is defined as the second largest local maximum of SDFs. To get quantitative information about the crossing fibers, we found the ratio of the number of F_{d2} for f_{32}

(or f_{cm}) to the number of F_{d2} for f_{64} .

The calculated results in the three ROIs (red, green, and blue boxes marked in Fig. 3(b)) were summarized in Table 3. The means (standard deviations) of the angles between fibers are calculated in each ROI. We observed that f_{cm} improved the fiber orientations and the quality of the crossing fibers. Although the mean of angles between the major fibers of f_{64} and f_{cm} was larger than that of f_{64} and f_{32} in green region, the ratio of the numbers of F_{d2} for f_{cm} and f_{64} was larger than that for f_{32} and f_{64} .

In Fig. 6(b), (c), and (d), we present the tractography generated by f_{64} , f_{32} , and f_{cm} , respectively, for comparison. The association fiber tracts that pass through the seeding regions marked in Fig. 6(a). Although most tracts show a roughly similar pathway, the reconstructed f_{cm} improved the quality of white matter tracts. In Fig. 6(b), we compared the qualities of tractography by f_{64} , f_{32} , and f_{cm} , respectively (see also Supplementary Fig. 4).

4.2. Enhancement of SDF resolution using HARDI data sets

Using the scanned HARDI data sets, Case 1 (64 DG directions) and Case 2 (32 DG directions), we reconstructed SDF maps with improved resolution by increasing the DG directions. The DG directions were extended to 128 and 64 for Case 1 and Case 2, respectively. We used the full measured diffusion MR signals $\{S(\mathbf{g}_1), \dots, S(\mathbf{g}_{64})\}$ to reconstruct new 64 diffusion MR signals, and denoted by f_{128} the corresponding SDFs for Case 1. For Case 2, new 32 DG directions were similarly synthesized using the proposed method and the corresponding new 32 diffusion MR signals were reconstructed.

Case 1: In Fig. 7(c)–(e), we compared the recovered SDFs, f_{64} and f_{128} , in the three ROIs (red, green and blue boxes) denoted in Fig. 7(a) and (b). In each panel, a representative SDF is enlarged for comparison. We observed that in nearly all voxels shown, f_{128} improved the shape of SDFs. The number of predicted DG directions was 158, 185, and 192 at the center of selected red, green, and blue regions, respectively. The sampling length ratios σ were 1 and 1.1 for f_{64} and f_{128} , respectively.

Case 2: First, we found SDFs, denoted by f_{32} , using the full measured 32 diffusion MR signals. Next, we used these diffusion MR signals to reconstruct new 32 diffusion MR signals and denoted by f_{cm} the corresponding SDFs. The sampling length ratios σ were 0.8 and 1 for f_{32} and f_{cm} , respectively.

In Fig. 8(c)–(e), we compared the recovered SDFs, f_{32} and f_{cm} , in the

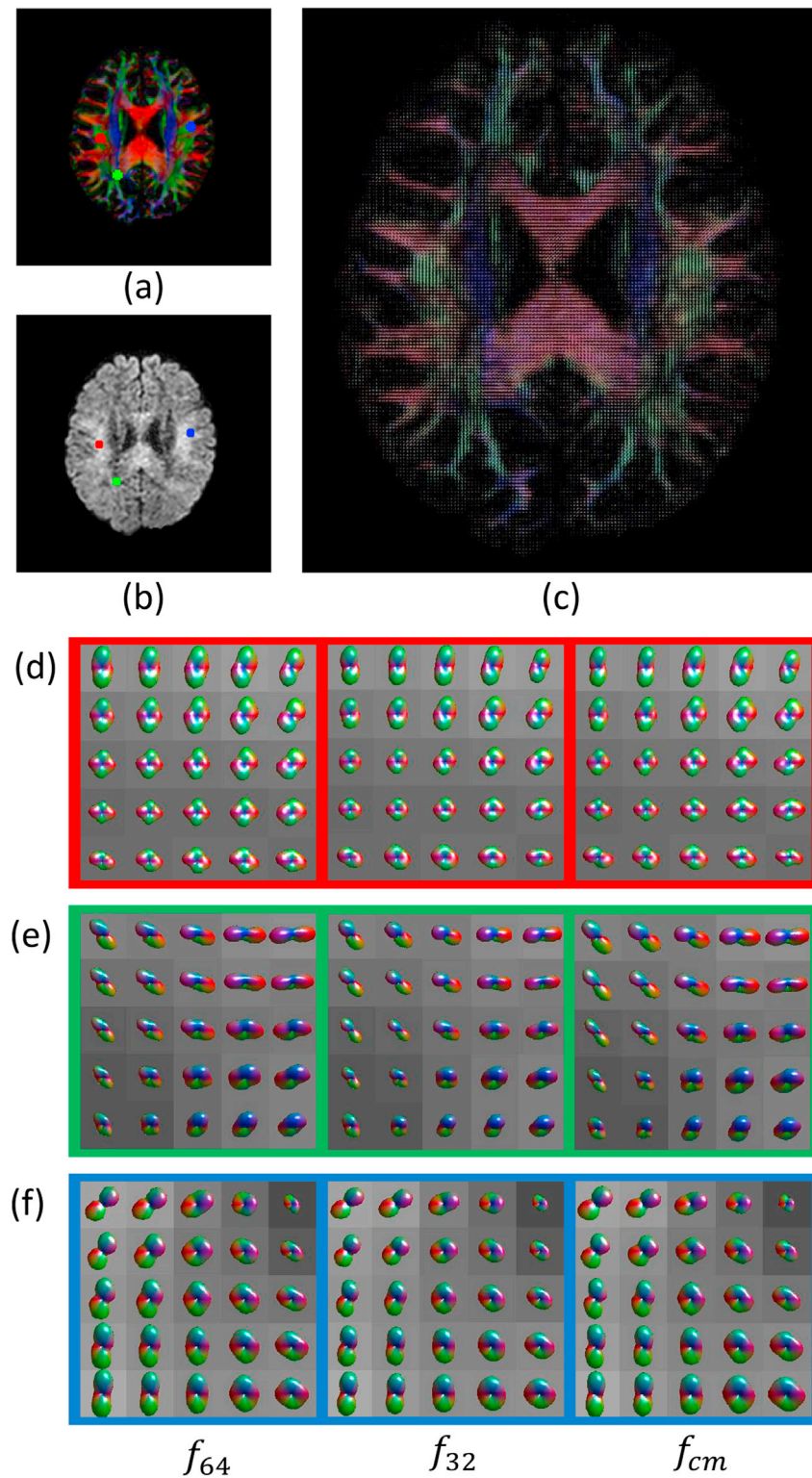


Fig. 3. (A) Color-coded fractional anisotropic (FA) map using the diffusion tensor. (b) Selected ROIs marked by red, green, and blue boxes, overlaid on the diffusion weighted image. (c) Recovered SDFs f_{cm} in the whole brain region. (d)–(f) Recovered SDFs f_{64} , f_{32} , and f_{cm} in the selected ROIs marked by red, green, and blue regions, respectively.

three ROIs (red, green and blue boxes) denoted in Fig. 8(a) and (b). In each panel, a representative SDF is enlarged for comparison. We observed that in nearly all voxels shown, f_{cm} improved the shape of SDFs.

The number of predicted DG directions was 73, 84, and 100 at the center of selected red, green, and blue regions, respectively.

5. Discussion

The widely used diffusion tensor model including 6-unknowns cannot determine the crossing fibers. The proposed approach for SDF aims to reduce exhaustive sampling numbers by generating diffusion MR signals in new predicted diffusion gradient directions. The proposed method

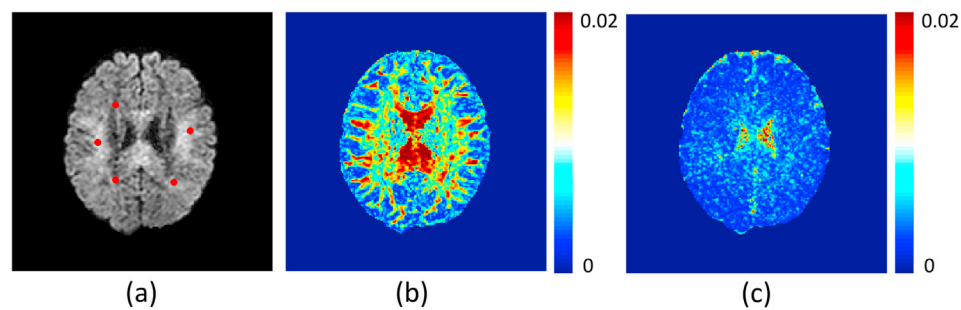


Fig. 4. (A) Voxel locations (red spots) to compare the quality of reconstructed SDFs in Table 1. (b) and (c) Relative L^2 -errors RE_{32} and RE_{cm} in (11) across all voxels in the brain region, respectively.

Table 1
Recovered SDFs f_{64} , f_{32} , and f_{cm} , respectively, with resolved fiber orientation of the corresponding SDFs and the relative L^2 -errors RE_{32} and RE_{cm} in (11).

f_{64}	f_{32}	f_{cm}	RE_{32}	RE_{cm}
			0.1080	0.0282
			0.0968	0.0324
			0.1150	0.0333
			0.0780	0.0229
			0.1091	0.0199

At most voxel locations, f_{cm} shows a profile closer to f_{64} than f_{32} .

leans on the estimation of likely diffusion MR signals in selected directions of each voxel based on the similarity of diffusion MR signals in neighboring directions. Under the assumption that anisotropic diffusion properties in the nervous system are strongly related each other correlated with specific cognitive functions, depending on the local diffusion patterns in a set of adjacent GD directions, an unmeasured (but reliable)

DG direction can be synthesized as an appropriate combination of adjacent DG directions. As a local diffusion pattern map of raw HARDI signals, we used a spatial local patch at each DG direction and predicted a new DG direction by the weighted combination of measured positions in the q -space on a single sphere. By synthesizing a new diffusion signal using the predicted DG direction, we can expect that the synthesized diffusion signal would be better than the typical interpolation method. We used a correlation coefficient between the behavior of diffusion signals in a local window as the distance of local diffusion pattern. The difference between the synthesized diffusion signal and the unmeasured actual diffusion signal in the new DG direction depends on the local diffusion pattern and measured noise. The noise variance of the synthesized diffusion signal in the new direction is less than the noise level of diffusion signals at each measured position on the sphere. The synthesized diffusion signal at the predicted position on the sphere can approximately recover the actual non-measured diffusion signal by analyzing the local diffusion pattern map.

In this method, we use a correlation value between the behavior of diffusion signals in a local window as a measure of the local diffusion pattern. We chose the regularization parameter λ in (8) based on GDV. It would be useful to find an algorithm for a “good” choice of λ for a given SDF based on neighboring behavior, or across all SDFs.

The proposed method has some limitations to overcome. First, the determined diffusion MR signal $S(\mathbf{u}_c) = \sum_{k=1}^K w_k S(\mathbf{g}_k)$ in (5) is a convex combination of measured diffusion MR signals in an adjacent neighborhood. For this reason, the strength of predicted diffusion MR signal cannot be greater than the intensity of diffusion MR signals in the adjacent neighborhood $\mathcal{N}_{\mathbf{v}_j}^K$. However, depending on the axonal distribution,

Table 2
Relative L^2 -errors, RE_{32} and RE_{cm} , in the different ROIs in Fig. 3(b).

RE_{32} in red region					RE_{cm} in red region				
0.116	0.092	0.090	0.109	0.132	0.034	0.052	0.039	0.032	0.053
0.094	0.085	0.089	0.109	0.117	0.048	0.051	0.033	0.040	0.050
0.092	0.085	0.097	0.114	0.103	0.029	0.040	0.032	0.037	0.028
0.114	0.121	0.101	0.091	0.110	0.024	0.023	0.029	0.060	0.047
0.088	0.096	0.100	0.106	0.126	0.034	0.023	0.022	0.036	0.060
RE_{32} in green region					RE_{cm} in green region				
0.120	0.092	0.106	0.147	0.157	0.060	0.070	0.045	0.038	0.040
0.109	0.084	0.104	0.149	0.140	0.090	0.088	0.038	0.031	0.022
0.095	0.085	0.115	0.140	0.113	0.068	0.063	0.033	0.053	0.014
0.075	0.083	0.095	0.118	0.143	0.022	0.036	0.065	0.034	0.037
0.065	0.071	0.098	0.109	0.110	0.031	0.050	0.060	0.043	0.028
RE_{32} in blue region					RE_{cm} in blue region				
0.103	0.109	0.106	0.081	0.075	0.022	0.022	0.026	0.023	0.016
0.124	0.116	0.098	0.083	0.083	0.036	0.041	0.021	0.023	0.026
0.134	0.122	0.109	0.091	0.106	0.031	0.025	0.020	0.019	0.030
0.147	0.139	0.127	0.112	0.102	0.039	0.031	0.021	0.024	0.016
0.152	0.144	0.130	0.110	0.087	0.037	0.026	0.021	0.016	0.029

The actual relative errors at each voxel location in the different ROIs are shown in Table 2.

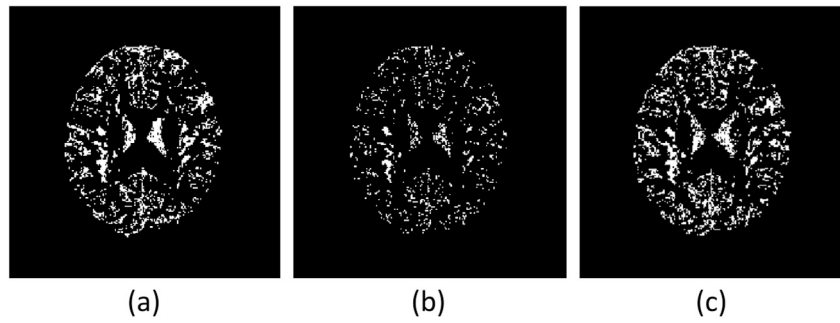


Fig. 5. Images representing the locations where the number of crossing fibers is larger than one, corresponding to (a) f_{64} , (b) f_{32} , and (c) f_{cm} , respectively.

Table 3

Summary of the results of resolved fiber structures in the different ROIs marked in Fig. 3(b). The mean (standard deviation) of the angles are calculated in each ROI.

		Red region	Green region	Blue region
Angle between major fibers	f_{64} and f_{32}	3.72 ° (4.30°)	5.08 ° (4.72°)	5.06 ° (4.85°)
	f_{64} and f_{cm}	3.00 ° (4.09°)	3.73 ° (4.85°)	5.71 ° (5.88°)
	f_{32} and f_{cm}			
Ratio of the number of F_{d2}	f_{64} and f_{32}	40%	14%	50%
	f_{64} and f_{cm}	73%	71%	100%
	f_{32} and f_{cm}			

the water molecule diffusivity can be high at an interior position of the neighborhood $\mathcal{N}_{v_j}^K$. As a possible solution to this problem, if a maximum value of diffusivity occurs in the interior position of the neighborhood, it is necessary to observe the local diffusion map $p_n(S(\mathbf{g}_{jm}))$ in (1) and to utilize *a priori* information to estimate the inside diffusivity information from the surrounding measured signals. Second, we used a rectangular region and divided it four triangular regions as neighborhood $\mathcal{N}_{v_j}^K$ to predict a DG direction. The synthesized diffusion signal at the predicted direction in the neighborhood $\mathcal{N}_{v_j}^K$ depends on the similarity patterns, which is a geometric constraint. For these reasons, the use of uniformly distributed measured DG directions can be expected to provide a good

approximation $S(\sum_{k=1}^K w_k \mathbf{g}_{jk}) \approx \sum_{k=1}^K w_k S(\mathbf{g}_{jk})$.

We focus on single shell SDFs in this paper. The method can be expandable to multi-shell acquisitions or DSI-style acquisitions. Since NODDI requires at least two HARDI shells with different b -value for estimating the neurite morphology, the technique requires relatively long acquisition times and enforces trade-offs in the reconstructed image quality ((Zhan et al., 2008)). Acquiring images with higher resolution increase the precision of anisotropy, but there is a trade-off between angular resolution and MR scan time. The proposed method can be easily extended to DWI using q -space data. We simply applied the proposed method to NODDI (see Supplementary Figs. 5 and 6). For NODDI, the diffusion MR data consist of three HARDI shells with $b = 2400 \text{ s/mm}^2$ (60 noncollinear gradient directions and six $b = 0$ images), $b = 800 \text{ s/mm}^2$ (30 noncollinear gradient directions and three $b = 0$ images) and $b = 300 \text{ s/mm}^2$ (9 noncollinear gradient directions and one $b = 0$ image). To validate the proposed method, we increased the angular resolution for the b -values ($b = 800 \text{ s/mm}^2$ and $b = 2400 \text{ s/mm}^2$). For the multi-shell acquisitions or DSI-style acquisitions, it will be a promising work to develop a method to predict the best location combining multi-shell q -space data simultaneously.

We need to develop an appropriate pattern map to determine the best positions for SDFs. The compressed sensing technique has a potential to provide HARDI using a reduced number of diffusion encoding gradients ((Michailovich and Rathi, 2010; Michailovich et al., 2011; Kuhnt et al., 2013)). We expect the proposed method combined with the compressed sensing technique can further reduce the MR scanning time.

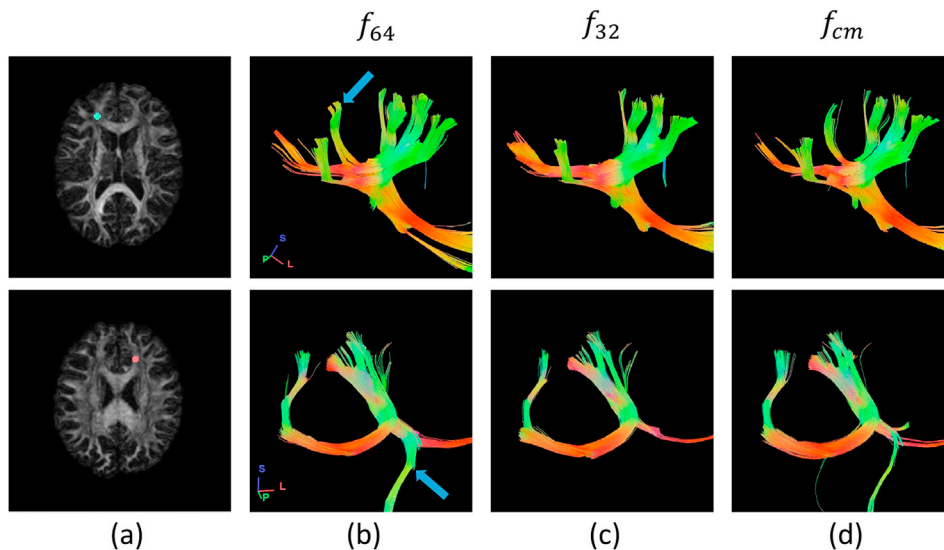


Fig. 6. (A) Seed regions for tractography are marked. (b), (c), and (d) Tractography generated by f_{64} , f_{32} , and f_{cm} , respectively.

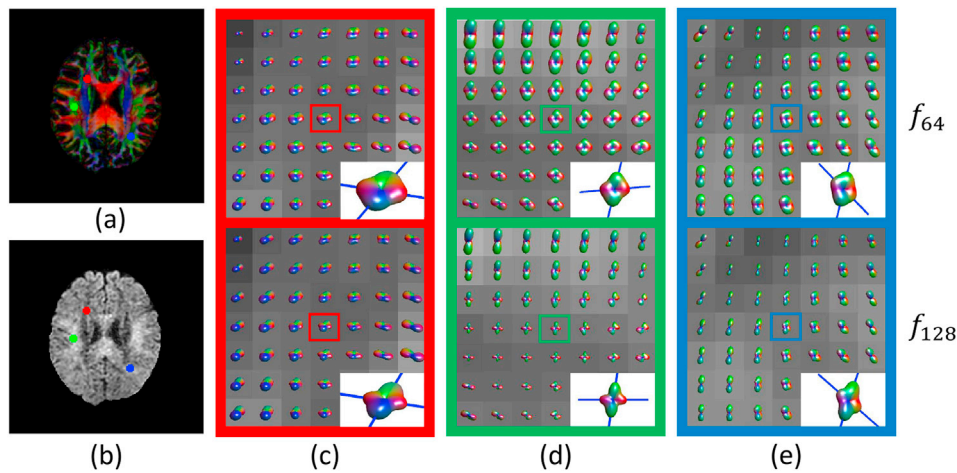


Fig. 7. (A) Color-coded fractional anisotropic (FA) map using the diffusion tensor. (b) Selected ROIs marked by red, green, and blue boxes are overlaid on the diffusion weighted image. (c) Recovered SDFs f_{64} (upper) and f_{128} (lower) in the red region. (d) Recovered SDFs f_{64} (upper) and f_{128} (lower) in the green region. (f) Recovered SDFs f_{64} (upper) and f_{128} (lower) in the blue region. A representative SDF is enlarged for comparison on each panel.

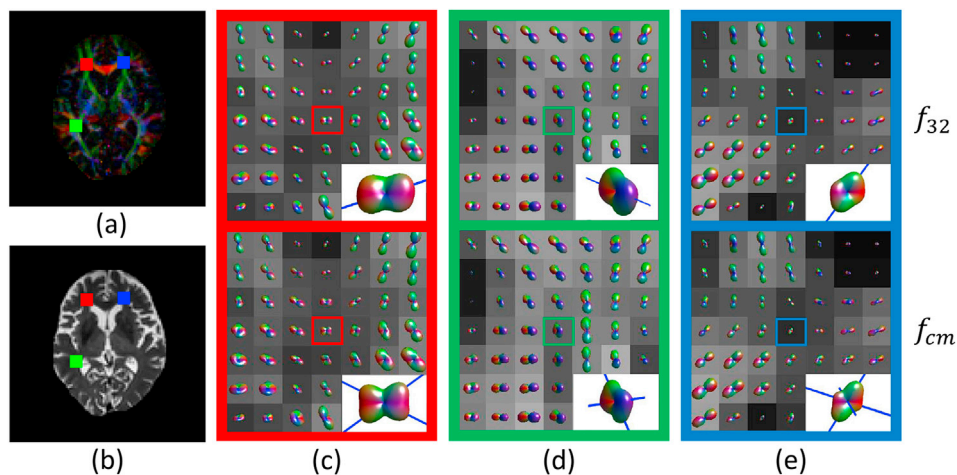


Fig. 8. (A) Color-coded fractional anisotropic (FA) map using the diffusion tensor. (b) Selected ROIs marked by red, green, and blue boxes overlaid on the diffusion weighted image. (c) Recovered SDFs f_{32} (upper) and f_{cm} (lower) in the red region. (d) Recovered SDFs f_{32} (upper) and f_{cm} (lower) in the green region. (e) Recovered SDFs f_{32} (upper) and f_{cm} (lower) in the blue region. In each panel, a representative SDF is enlarged for comparison.

6. Conclusion

A long acquisition time is a major drawback of HARDI in clinical implementation. The developed method provides an enhanced angular resolution of DWI by determining appropriate diffusion gradient directions, at which a synthesized signal is obtained by combining surrounding diffusion MR signals in given diffusion gradient directions. We propose a method to address the interpolation of diffusion MR data, with an ultimate goal of reducing acquisition time for diffusion MRI, while enhancing the angular resolution of DWI. The proposed method can increase the quality of underlying fiber orientation map for analyzing diffusivity properties without increasing MR scan time (or with using already scanned MR data). We use a local diffusion pattern map of diffusion MR single- or multi-shell acquisitions, which determines appropriate diffusion gradient directions to be synthesized within a nearest neighborhood diffusion gradient group.

The predicted DG directions are chosen to balance the similarity between members of a neighborhood group and experimentally reduce the error between the synthesized values and the true values. Using the synthesized and measured diffusion signals, we estimated the SDF map with human brain data. We validated the proposed method by comparing the recovered SDFs obtained from the synthesized and measured signals

in 32 DG directions (denoted f_{cm}) with the recovered SDFs obtained from only the measured signals in 32 DG directions (denoted f_{32}). We observed that when including synthesized diffusion signals, the SDFs f_{cm} were close to the SDFs obtained from the full measured data in 64 DG directions (denoted f_{64}). We also observed that the SDFs f_{cm} generated a multiple fiber structure which more closely matches the ODFs f_{64} . Significantly, in some crossing fibers, f_{cm} showed the same directional pattern as in f_{64} , while f_{32} showed the opposite directional pattern.

Acknowledgment

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.neuroimage.2018.08.072>.

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